

FMLG-TB Tubular Level Indicator

Side-Mounted Gauge Glass

The Tubular Level Indicator (side mounted) is a reliable device for reading the level of liquid in a tank. It is the simplest form of level indicator and finds wide application in process tanks, batching tanks, and similar vessels.

General Features

- Gauge fitted between two end blocks through gland packings, forming a closed loop with the tank.
- Guards provided as tie-rods / C-channels around the gauge to protect against accidental knocks.
- End blocks have built-in isolating valve, drain valve, and vent plug.

Industries & Applications

- Chemical / Process Industry: Process tanks, batching tanks, and bulk storage tanks.
- Marine / Shipping: Marine tank gauging (oil/water tanks on ships and barges).
- Water Treatment / Dosing: Seal pots and calibration pots for dosing-tank flow verification.
- Fertilizer: Liquid fertilizer mixing stations.
- Chemical Storage: Non-corrosive raw material storage vessels.



Key Specifications

Operating Pressure	Up to 15 kg/cm ²
Operating Temperature	Up to 100C
C-C Distance	Up to 4000 mm
Mounting	Side

General Specifications — FMLG-TB (Side-Mounted Gauge Glass)

Parameter	Specification
Model	FMLG-TB
Mounting Type	Side
Glass Tube	Heavy-walled Borosilicate Glass, H-19 / 16 Teflon
End Block MOC	CS / SS304 / SS316 / PP / PVDF
Protective Channel	CS rectangular channel box, powder coated
Packing	Teflon
Connection	Flanged ASA
Valve MOC	CS / SS304 / SS316 / PP / PVDF
Process Connection	Flanged or Screwed
Operating Pressure	Up to 15 kg/cm ²
Operating Temperature	Up to 100C

Parameter	Specification
C-C Distance (+/-3mm)	Up to 4000 mm (beyond 4000 mm, consult factory)
Visibility	C-C distance minus 150 mm

Ordering Code

Supplied up to 3500 mm in single-piece design; above 3500 mm, consult factory.

Segment	Parameter	Options	Digit / Code
1	Mounting	Side	SD
2	Body Material	Carbon Steel	CS
		AISI 304 SS Channel	4S
		AISI 316 SS Channel	SS
		Non Standard	XX
3	Process Connection	1/2" NPT (M) / (F)	5NM / 5NF
		3/4" NPT (M) / (F)	7NM / 7NF
		1" NPT (M) / (F)	1NM / 1NF
		1/2" BSP (M) / (F)	5BM / 5BF
		3/4" BSP (M) / (F)	7BM / 7BF
		1" BSP (M) / (F)	1BM / 1BF
		Flanged 1/2"-3" (150#/300#/600# RF, ANSI B16.5 — see Flange Code Reference below)	F501-F306
		Non Standard	XX
4	Top Side Vent	Plug - 1/2" NPT (M)	V1
		Plug - 3/4" NPT (M)	V2
		Plug - 1" NPT (M)	V3
		Valve - 1/2" NPT (F)	V4
		Valve - 3/4" NPT (F)	V5
		Flanged (see Flange Code Reference below)	F501-F306
		Non Standard	XX
5	Bottom Side Drain	Plug - 1/2" NPT (M)	V1
		Plug - 3/4" NPT (M)	V2
		Plug - 1" NPT (M)	V3
		Valve - 1/2" NPT (F)	V4
		Valve - 3/4" NPT (F)	V5
		Flanged (see Flange Code Reference below)	F501-F306
		Non Standard	XX
6	Material of Scale	Aluminium (Standard)	AL
		AISI 304 SS	4S
7	Center to Center Distance	Specify in mm, e.g. 800 for 800 mm	L####

Segment	Parameter	Options	Digit / Code
8	Options	Ball Valve - Bolted Bonnet Type	BB
		Non Standard	XX

Ordering example (illustrative): FMLG-TB-SD-4S-F501-V1-V2-AL-L0800

Modifications may take place and materials specified may be replaced without prior notice. Confirm all fields before placing an order.

Flange Code Reference (ANSI B16.5, Raised Face)

Size	Rating	Code	Size	Rating	Code
1/2"	150#	F501	1-1/2"	150#	F151
1/2"	300#	F503	1-1/2"	300#	F153
1/2"	600#	F506	1-1/2"	600#	F156
3/4"	150#	F751	2"	150#	F201
3/4"	300#	F753	2"	300#	F203
3/4"	600#	F756	2"	600#	F206
1"	150#	F101	3"	150#	F301
1"	300#	F103	3"	300#	F303
1"	600#	F106	3"	600#	F306

FMLG-MG Magnetic Level Indicator (Capsule Type)

Bypass Gauge

The magnetic bypass level indicator is installed outside a vessel or tank. Liquid level is read from the change of flag colour as a magnetic float rises and falls with the process level. Optional magnetic switches and a level transmitter can be added for electrical signal output.

General Features

- Float with internal magnet rises and drops with liquid level; magnetic flags flip to indicate level.
- Optional magnetic level switches, mountable anywhere along the indication rail.
- Optional 4-20 mA magnetic level transmitter, accuracy +/-3mm.

Industries & Applications

- Chemical Processing: Hazardous chemical reactors.
- Fertilizer: Ammonia storage tanks.
- Chemical / Acid Handling: Concentrated acid and corrosive-media tanks (glandless, non-contact float).
- Petrochemical: Toxic solvent processing lines.
- Power Generation: Boiler feedwater tanks (preferred over gauge glass by design code).



Key Specifications

Operating Pressure	Up to 50 kg/cm ²
Operating Temperature	-20 to 400C (with cooling fins)
C-C Distance	Up to 6000 mm
Measuring Error	+/-10 mm

General Specifications — FMLG-MG (Bypass Gauge)

Parameter	Specification
Model	FMLG-MG
Mounting Type	Top or Side
Density (min.)	From 0.70 g/cm ³
Precision (interface/density)	From +/-0.01 g/cm ³
Measuring Error	+/-10 mm
Viscosity (max.)	1000 mPa.s
Connection	Weld End (standard) / Flanges (DIN and ANSI)
Weight	Basic 7 kg
Wetted Parts (Float/Chamber/Process Conn.)	AISI 316 SS or AISI 316L SS
Operating Pressure	Up to 50 kg/cm ²

Parameter	Specification
Operating Temperature	-20C to 120C (up to 400C with cooling fins)
Process Connection	Flanged or Screwed (screwed for side mounting only)
C-C Distance (+/-3mm)	Up to 6000 mm
Flag Display Length	Up to 6000 mm
Specific Gravity (min.)	0.75
Indicator Roller	Metallic

Ordering Code

Supplied up to 3500 mm in single-piece design; above 3500 mm, supplied in a two-piece coupled design. Screwed process connections apply to side mounting only.

Segment	Parameter	Options	Digit / Code
1	Mounting	Top	TP
		Side	SD
2	Wetted Parts	AISI 316 SS	SS
		AISI 316L SS	SL
3	Temperature Range	-20C ~ 120C	T1
		-20C ~ 400C (with cooling fins)	T2
4	Pressure Range	0-10 kg/cm ²	P1
		0-25 kg/cm ²	P2
		0-50 kg/cm ²	P3
5	Vent	Screwed cap + plug, 1/2"/3/4"/1" NPT (M)	V1 / V2 / V3
		Screwed cap + valve, 1/2"/3/4" NPT (F)	V4 / V5
		Single flanged (top mounted)	V6
		Welded flat top + plug, 1/2"/3/4"/1" NPT (M)	V7 / V8 / V9
		Welded flat top + valve, 1/2"/3/4" NPT (F)	V10 / V11
		Non Standard	XX
6	Drain	Dual flange + plug, 1/2"/3/4"/1" NPT (M)	D1 / D2 / D3
		Dual flange + valve, 1/2"/3/4" NPT (M)	D4 / D5
		Non Standard	XX
7	Process Connection	Screwed 1/2"-1" NPT/BSP (side mounting only)	5NM-1BF
		Flanged 1/2"-3" (150#/300#/600# RF — see Flange Code Reference below)	F501-F306
		Non Standard	XXX
8	Center to Center Distance	Specify in mm, e.g. 500 for 500 mm	L####
9	Material of Scale	Aluminium (Standard)	AL
		AISI 304 SS	4S
10	Flag Colour	Blue & White	BW

Segment	Parameter	Options	Digit / Code
		Red & White	RW
		Black & White	KW
11	Magnetic Switches (Optional)	One	1
		Two	2
		Three	3
12	Switch Housing (Optional)	SPST reed switch, general purpose housing	1
		SPDT micro switch, IP65 housing	2
		SPDT micro switch, Ex-proof housing	3

Ordering example (illustrative): FMLG-MG-SD-SS-T1-P3-V4-D1-F501-L1000-AL-RW-2-2

Modifications may take place and materials specified may be replaced without prior notice. Confirm all fields before placing an order.

Flange Code Reference (ANSI B16.5, Raised Face)

Size	Rating	Code	Size	Rating	Code
1/2"	150#	F501	1-1/2"	150#	F151
1/2"	300#	F503	1-1/2"	300#	F153
1/2"	600#	F506	1-1/2"	600#	F156
3/4"	150#	F751	2"	150#	F201
3/4"	300#	F753	2"	300#	F203
3/4"	600#	F756	2"	600#	F206
1"	150#	F101	3"	150#	F301
1"	300#	F103	3"	300#	F303
1"	600#	F106	3"	600#	F306

FMLG-BM Bi-Colour Magnetic Level Indicator

Bypass Gauge

The bi-colour magnetic bypass level indicator is installed outside a vessel or tank. Liquid level is read from the change of flag colour as a magnetic float rises and falls with the process level. Optional magnetic switches and a level transmitter can be added for electrical signal output.

General Features

- Float with internal magnet rises and drops with liquid level; magnetic flags flip to indicate level.
- Optional magnetic level switches, mountable anywhere along the indication rail.
- Optional 4-20 mA magnetic level transmitter, accuracy +/-3mm.

Industries & Applications

- Petrochemical / Refining: Top-tier distillation columns.
- Oil & Gas / LPG: Pressurized liquid gas storage bullets.
- Specialty Chemicals: Elevated chemical synthesis reactors.
- Chemical Processing: Interface and colour-change monitoring (bi-colour flag distinguishes liquid from vapour at a glance).



Key Specifications

Operating Pressure	Up to 50 kg/cm ²
Operating Temperature	-20 to 400C
C-C Distance	Up to 6000 mm
Measuring Error	+/-10 mm

General Specifications — FMLG-BM (Bypass Gauge)

Parameter	Specification
Model	FMLG-BM
Mounting Type	Top or Side
Density (min.)	From 0.70 g/cm ³
Precision (interface/density)	From +/-0.01 g/cm ³
Measuring Error	+/-10 mm
Viscosity (max.)	1000 mPa.s
Connection	Weld End (standard) / Flanges (DIN and ANSI)
Weight	Basic 7 kg
Wetted Parts (Float/Chamber/Process Conn.)	AISI 316 SS or AISI 316L SS
Operating Pressure	Up to 50 kg/cm ²

Parameter	Specification
Operating Temperature	-20C to 400C
Process Connection	Flanged or Screwed (screwed for side mounting only)
C-C Distance (+/-3mm)	Up to 6000 mm
Flag Display Length	Up to 6000 mm
Specific Gravity (min.)	0.75
Indicator Roller	Metallic

Ordering Code

Supplied up to 3500 mm in single-piece design; above 3500 mm, supplied in a two-piece coupled design. Screwed process connections apply to side mounting only.

Segment	Parameter	Options	Digit / Code
1	Mounting	Top	TP
		Side	SD
2	Wetted Parts	AISI 316 SS	SS
		AISI 316L SS	SL
3	Temperature Range	-20C ~ 120C	T1
		-20C ~ 400C (with cooling fins)	T2
4	Pressure Range	0-10 kg/cm ²	P1
		0-25 kg/cm ²	P2
		0-50 kg/cm ²	P3
5	Vent	Screwed cap + plug, 1/2"/3/4"/1" NPT (M)	V1 / V2 / V3
		Screwed cap + valve, 1/2"/3/4" NPT (F)	V4 / V5
		Single flanged (top mounted)	V6
		Welded flat top + plug, 1/2"/3/4"/1" NPT (M)	V7 / V8 / V9
		Welded flat top + valve, 1/2"/3/4" NPT (F)	V10 / V11
		Non Standard	XX
6	Drain	Dual flange + plug, 1/2"/3/4"/1" NPT (M)	D1 / D2 / D3
		Dual flange + valve, 1/2"/3/4" NPT (M)	D4 / D5
		Non Standard	XX
7	Process Connection	Screwed 1/2"-1" NPT/BSP (side mounting only)	5NM-1BF
		Flanged 1/2"-3" (150#/300#/600# RF — see Flange Code Reference below)	F501-F306
		Non Standard	XXX
8	Center to Center Distance	Specify in mm, e.g. 500 for 500 mm	L####
9	Material of Scale	Aluminium (Standard)	AL
		AISI 304 SS	4S
10	Flag Colour	Blue & White	BW

Segment	Parameter	Options	Digit / Code
		Red & White	RW
		Black & White	KW
11	Magnetic Switches (Optional)	One	1
		Two	2
		Three	3
12	Switch Housing (Optional)	SPST reed switch, general purpose housing	1
		SPDT micro switch, IP65 housing	2
		SPDT micro switch, Ex-proof housing	3

Ordering example (illustrative): FMLG-BM-TP-SL-T2-P2-V6-D2-F103-L2000-4S-BW-1-1

Modifications may take place and materials specified may be replaced without prior notice. Confirm all fields before placing an order.

Flange Code Reference (ANSI B16.5, Raised Face)

Size	Rating	Code	Size	Rating	Code
1/2"	150#	F501	1-1/2"	150#	F151
1/2"	300#	F503	1-1/2"	300#	F153
1/2"	600#	F506	1-1/2"	600#	F156
3/4"	150#	F751	2"	150#	F201
3/4"	300#	F753	2"	300#	F203
3/4"	600#	F756	2"	600#	F206
1"	150#	F101	3"	150#	F301
1"	300#	F103	3"	300#	F303
1"	600#	F106	3"	600#	F306

FMLG-TM Top-Mounted Magnetic Level Indicator

Bypass Gauge

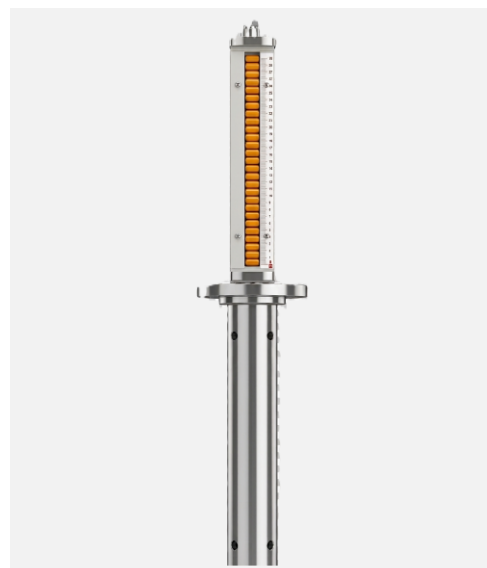
The top-mounted magnetic bypass level indicator is installed outside a vessel or tank. Liquid level is read from the change of flag colour as a magnetic float rises and falls with the process level. Optional magnetic switches and a level transmitter can be added for electrical signal output.

General Features

- Float with internal magnet rises and drops with liquid level; magnetic flags flip to indicate level.
- Optional magnetic level switches, mountable anywhere along the indication rail.
- Optional 4-20 mA magnetic level transmitter, accuracy +/-3mm.

Industries & Applications

- Chemical / Hazardous Waste: Subsurface hazardous chemical waste collection pits.
- Environmental / Effluent Treatment: Neutralization sumps.
- Industrial Facilities: Floor-drain containment tanks.
- Chemical Processing: Underground or sump tanks with no side access for mounting.



Key Specifications

Operating Pressure	Up to 50 kg/cm ²
Operating Temperature	-20 to 400C
C-C Distance	Up to 6000 mm
Mounting	Top only

General Specifications — FMLG-TM (Bypass Gauge)

Parameter	Specification
Model	FMLG-TM
Mounting Type	Top
Density (min.)	From 0.70 g/cm ³
Precision (interface/density)	From +/-0.01 g/cm ³
Measuring Error	+/-10 mm
Viscosity (max.)	1000 mPa.s
Connection	Weld End (standard) / Flanges (DIN and ANSI)
Weight	Basic 7 kg
Wetted Parts (Float/Chamber/Process Conn.)	AISI 316 SS or AISI 316L SS
Operating Pressure	Up to 50 kg/cm ²
Operating Temperature	-20C to 400C

Parameter	Specification
Process Connection	Flanged (screwed connections apply to side-mounted variants only)
C-C Distance (+/-3mm)	Up to 6000 mm
Flag Display Length	Up to 6000 mm
Specific Gravity (min.)	0.75
Indicator Roller	Metallic

Ordering Code

Supplied up to 3500 mm in single-piece design; above 3500 mm, supplied in a two-piece coupled design.

Segment	Parameter	Options	Digit / Code
1	Wetted Parts	AISI 316 SS	SS
		AISI 316L SS	SL
2	Temperature Range	-20C ~ 120C	T1
		-20C ~ 400C (with cooling fins)	T2
3	Pressure Range	0-10 kg/cm2	P1
		0-25 kg/cm2	P2
		0-50 kg/cm2	P3
4	Vent	Screwed cap + plug, 1/2"/3/4"/1" NPT (M)	V1 / V2 / V3
		Screwed cap + valve, 1/2"/3/4" NPT (F)	V4 / V5
		Single flanged (top mounted)	V6
		Welded flat top + plug, 1/2"/3/4"/1" NPT (M)	V7 / V8 / V9
		Welded flat top + valve, 1/2"/3/4" NPT (F)	V10 / V11
		Non Standard	XX
5	Process Connection	Screwed 1/2"-1" NPT/BSP	5NM-1BF
		Flanged 1/2"-3" (150#/300#/600# RF — see Flange Code Reference below)	F501-F306
		Non Standard	XXX
6	Center to Center Distance	Specify in mm, e.g. 500 for 500 mm	L#####
7	Material of Scale	Aluminium (Standard)	AL
		AISI 304 SS	4S
8	Flag Colour	Blue & White	BW
		Red & White	RW
		Black & White	KW
9	Magnetic Switches (Optional)	One	1
		Two	2
		Three	3

Segment	Parameter	Options	Digit / Code
10	Switch Housing (Optional)	SPST reed switch, general purpose housing	1
		SPDT micro switch, IP65 housing	2
		SPDT micro switch, Ex-proof housing	3

Ordering example (illustrative): FMLG-TM-SS-T1-P1-V6-F501-L1500-AL-KW-1-1

Modifications may take place and materials specified may be replaced without prior notice. Confirm all fields before placing an order.

Flange Code Reference (ANSI B16.5, Raised Face)

Size	Rating	Code	Size	Rating	Code
1/2"	150#	F501	1-1/2"	150#	F151
1/2"	300#	F503	1-1/2"	300#	F153
1/2"	600#	F506	1-1/2"	600#	F156
3/4"	150#	F751	2"	150#	F201
3/4"	300#	F753	2"	300#	F203
3/4"	600#	F756	2"	600#	F206
1"	150#	F101	3"	150#	F301
1"	300#	F103	3"	300#	F303
1"	600#	F106	3"	600#	F306

FMLG-RF Reflex Level Indicator

High-Pressure Visual Gauge

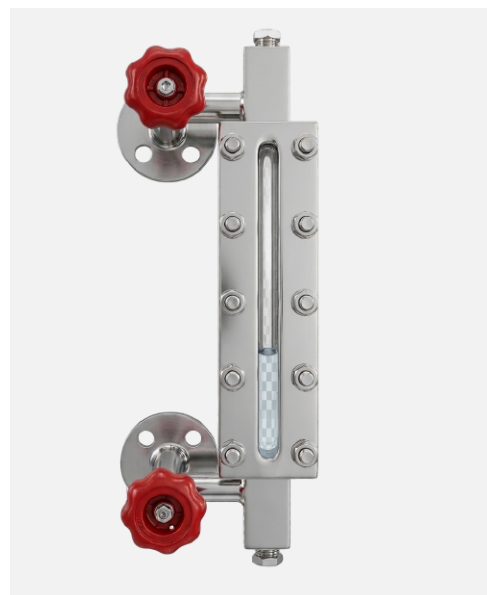
Reflex level indicators are designed for safe, positive visual indication of liquid level in vessels under high pressure and temperature. Precision-moulded prismatic grooves on the inner glass surface reflect or absorb light depending on whether the glass is covered by liquid or vapour, marking a sharp reading line.

General Features

- Liquid-covered glass appears black; vapour-space glass appears silvery-white.
- Offset needle valve x auto ball check x screwed/union/bolted bonnet isolation valves.
- Optional frost-free extension and Ex-proof / weatherproof jacketing.

Industries & Applications

- Power Generation: Steam boiler drums (water-level monitoring to prevent dry-firing or overfilling).
- Petrochemical: Clean, colourless liquid service — water and hydrocarbon reactors (not suited to interface reading).
- Chemical Processing: Solvent distillation columns.
- Fertilizer: Pressurized liquid ammonia storage.
- Oil & Gas: Gas-liquefaction plants and high-pressure process vessels.



Key Specifications

Low Pressure	30 kg/cm ²
Medium Pressure	65 kg/cm ²
High Pressure	100 kg/cm ²
Isolation Valve (max.)	Bolted bonnet, 210 kg/cm ²

General Specifications — FMLG-RF (High-Pressure Visual Gauge)

Parameter	Specification
Model	FMLG-RF
Mounting Type	Side
Gauge Classification	Low 30 kg/cm ² Medium 65 kg/cm ² High 100 kg/cm ²
Gauge Glass	Tempered soda-ash/Borosilicate (30W x 17mm) or Tempered Borosilicate (34W x 17mm)
Cushion / Gasket	CAF, CNAF, PTFE, SS304 or SS316 spiral-wound with graphite filler
Body (Liquid Chamber)	CS, ASTM A-105, SS304, SS316, or PP (CS reinforced)
Cover Plate	CS, ASTM A-105, SS304, SS316, or FRP
Bolts / Nuts	CS, SS304, or A193 Gr. B7
Process Connection Orientation	Rear, Rear-or-Left, Left-or-Right, Right, or Vertical
Isolation Valves	Offset needle valve x auto ball check x screwed bonnet (85 kg/cm ²) / union bonnet (165 kg/cm ²) / bolted bonnet (210 kg/cm ²)

Parameter	Specification
Vent / Drain	1/2" NPT (BSP for PP/Teflon MOC) plug or valve (ball, needle, diaphragm, globe, gate)
Calibrated Scale	Polycarbonate, SS304, or Aluminium
Special Features	Frost-free extension (30mm Perspex shield); jacketing to IP65 or Ex-proof Gr. IIA/IIB/IIC

Ordering Code

Longer C-C distances are provided as two chambers coupled with a flange coupler. Screwed connections use BSP threads for PP/Teflon MOC. Cushion/gasket, isolation-valve bonnet, and connection-orientation segments below are itemised as separate codes here so an order can be fully specified from this document alone.

Segment	Parameter	Options	Digit / Code
1	Body Material	Carbon Steel (rectangular channel box, powder coated)	C
		AISI 304 SS Channel	S
		AISI 316 SS Channel	T
2	Tube Guard	Carbon Steel	C
		AISI 304 SS	S
		AISI 316 SS	T
3	Cushion / Gasket	CAF	G1
		CNAF	G2
		PTFE	G3
		SS304 spiral-wound with graphite filler	G4
		SS316 spiral-wound with graphite filler	G5
4	Isolation Valve Bonnet	Screwed bonnet (max. 85 kg/cm ²)	B1
		Union bonnet (max. 165 kg/cm ²)	B2
		Bolted bonnet (max. 210 kg/cm ²)	B3
5	Process Connection Orientation	Rear	OR
		Rear or Left	OL
		Left or Right	OS
		Right	OD
		Vertical	OV
6	Process Connection	Screwed 1/2"-1" NPT/BSP	5NM-1BF
		Flanged 1/2"-3" (150#/300#/600# RF — see Flange Code Reference below)	F501-F306
		Non Standard	XXX
7	Top Side Vent	Plug 1/2"/3/4"/1" NPT (M)	PN1 / PN2 / PN3
		Needle Valve 1/2" NPT (F)	NV4
		Ball Valve 1/2" NPT (F)	Bv5

Segment	Parameter	Options	Digit / Code
		Flanged (see Flange Code Reference below)	F501-F306
		Non Standard	XXX
8	Bottom Side Drain	Plug 1/2"/3/4"/1" NPT (M)	PN1 / PN2 / PN3
		Needle Valve 1/2" NPT (F)	NV4
		Ball Valve 1/2" NPT (F)	Bv5
		Flanged (see Flange Code Reference below)	F501-F306
		Non Standard	XXX
9	Material of Scale	Aluminium (Standard)	AL
		AISI 304 SS	4S
10	Center to Center Distance	Specify in mm; supplied 350-3500mm single piece, above 3500mm two-piece coupled	L#####
11	Options	Ball Valve - Bolted Bonnet Type	BV

Ordering example (illustrative): FMLG-RF-T-T-G5-B3-OR-F501-PN1-PN1-AL-L0800-BV

Modifications may take place and materials specified may be replaced without prior notice. Confirm all fields before placing an order.

Flange Code Reference (ANSI B16.5, Raised Face)

Size	Rating	Code	Size	Rating	Code
1/2"	150#	F501	1-1/2"	150#	F151
1/2"	300#	F503	1-1/2"	300#	F153
1/2"	600#	F506	1-1/2"	600#	F156
3/4"	150#	F751	2"	150#	F201
3/4"	300#	F753	2"	300#	F203
3/4"	600#	F756	2"	600#	F206
1"	150#	F101	3"	150#	F301
1"	300#	F103	3"	300#	F303
1"	600#	F106	3"	600#	F306

FMLG-TP Transparent Level Indicator

High-Pressure Visual Gauge

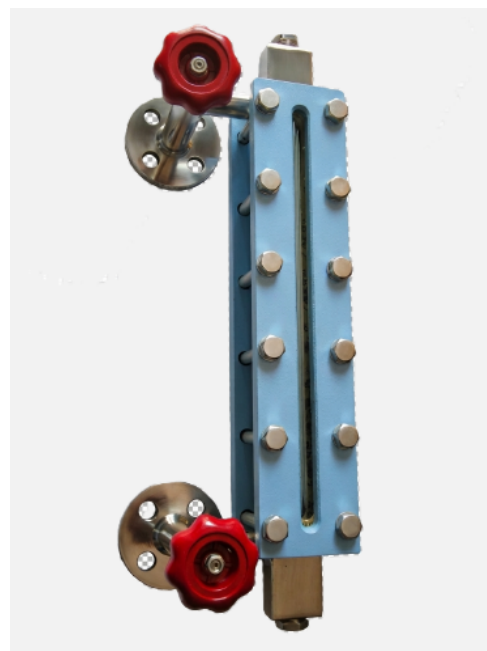
Transparent level indicators are designed for safe, positive visual indication of liquid level in vessels under high pressure and temperature. A clear glass suits interface services or dirty/viscous liquids where a reflex gauge is unsuitable.

General Features

- Liquid-covered glass appears black; vapour-space glass appears silvery-white.
- Suited to interface tracking and dirty or viscous media.
- Optional anti-frost extension and illuminator (weatherproof or Ex-proof).

Industries & Applications

- Chemical Processing: Interface tracking for multi-phase chemical reactors.
- Fertilizer / Agrochemicals: Colour-changing acid batching.
- Fertilizer: Liquid fertilizer blending vats.
- Chemical Processing: Dirty, viscous, or corrosive process media (with optional mica/Kel-F shields).
- Power Generation / Water Treatment: Treated-water, boiler, and condensate service.



Key Specifications

Low Pressure	30 kg/cm ²
Medium Pressure	65 kg/cm ²
High Pressure	100 kg/cm ²
Sight Glass	Borosilicate / Ilmadur / Klinger

General Specifications — FMLG-TP (High-Pressure Visual Gauge)

Parameter	Specification
Model	FMLG-TP
Mounting Type	Side
Gauge Classification	Low 30 kg/cm ² Medium 65 kg/cm ² High 100 kg/cm ²
Gauge Glass	Tempered soda-ash/Borosilicate (30W x 17mm) or Tempered Borosilicate (34W x 17mm)
Cushion / Gasket	CAF, CNAF, PTFE, SS304 or SS316 spiral-wound with graphite filler
Body (Liquid Chamber)	CS, ASTM A-105, SS304, SS316, or PP (CS reinforced)
Cover Plate	CS, ASTM A-105, SS304, SS316, or FRP
Bolts / Nuts	CS, SS304, or A193 Gr. B7
Process Connection Orientation	Rear, Rear-or-Left, Left-or-Right, Right, or Vertical

Parameter	Specification
Isolation Valves	Offset needle valve x auto ball check x screwed bonnet (85 kg/cm ²) / union bonnet (165 kg/cm ²) / bolted bonnet (210 kg/cm ²)
Vent / Drain	1/2" NPT (BSP for PP/Teflon MOC) plug or valve (ball, needle, diaphragm, globe, gate)
Calibrated Scale	Polycarbonate, SS304, or Aluminium
Special Features	Frost-free extension; jacketing to IP65/Ex-proof; illuminator (15W bulb, 230VAC, on 20mm Perspex shield)

Ordering Code

Longer C-C distances are provided as two chambers coupled with a flange coupler. Isolation-valve bonnet and connection-orientation segments below are itemised as separate codes here so an order can be fully specified from this document alone.

Segment	Parameter	Options	Digit / Code
1	Body Material	Carbon Steel	CS
		AISI 304 SS	4S
		AISI 316 SS	6T
2	Material of Cover Plates	Carbon Steel	CS
		AISI 304 SS	4S
		AISI 316 SS	6T
3	Sight Glass	Borosilicate Glass (max. 20 kg/cm ² , 120C)	BO
		Ilmadur (max. 20 kg/cm ² , 120C)	IL
		Klinger (max. 20 kg/cm ² , 120C)	KL
4	Gasket	CAF	CA
		Graphoil	GR
		PTFE	PT
5	Isolation Valve Bonnet	Screwed bonnet (max. 85 kg/cm ²)	B1
		Union bonnet (max. 165 kg/cm ²)	B2
		Bolted bonnet (max. 210 kg/cm ²)	B3
6	Process Connection Orientation	Rear	OR
		Rear or Left	OL
		Left or Right	OS
		Right	OD
		Vertical	OV
7	Process Connection	Screwed 1/2"-1" NPT/BSP	5NM-1BF
		Flanged 1/2"-3" (150#/300#/600# RF — see Flange Code Reference below)	F501-F306
		Non Standard	XXX

Segment	Parameter	Options	Digit / Code
8	Top Side Vent	Plug 1/2"/3/4"/1" NPT (M)	PN1 / PN2 / PN3
		Needle Valve 1/2" NPT (F)	NV4
		Ball Valve 1/2" NPT (F)	Bv5
		Flanged (see Flange Code Reference below)	F501-F306
		Non Standard	XXX
9	Bottom Side Drain	Plug 1/2"/3/4"/1" NPT (M)	PN1 / PN2 / PN3
		Needle Valve 1/2" NPT (F)	NV4
		Ball Valve 1/2" NPT (F)	Bv5
		Flanged (see Flange Code Reference below)	F501-F306
		Non Standard	XXX
10	Material of Scale	Aluminium (Standard)	AL
		AISI 304 SS	4S
11	Center to Center Distance	Specify in mm; supplied 350-3500mm single piece, above 3500mm two-piece coupled	L####
12	Options	With Anti Frost	WF
		Illuminator w/ Diffuser (Weatherproof IP65)	IW
		Illuminator w/ Diffuser (Ex-proof IIA/IIB + IP65)	IE
		Illuminator w/ Diffuser (Ex-proof IIC + IP65)	IX
		Ball Valve - Bolted Bonnet Type	BV

Ordering example (illustrative): FMLG-TP-6T-6T-BO-PT-B3-OR-F501-PN1-PN1-AL-L0800-IW

Modifications may take place and materials specified may be replaced without prior notice. Confirm all fields before placing an order.

Flange Code Reference (ANSI B16.5, Raised Face)

Size	Rating	Code	Size	Rating	Code
1/2"	150#	F501	1-1/2"	150#	F151
1/2"	300#	F503	1-1/2"	300#	F153
1/2"	600#	F506	1-1/2"	600#	F156
3/4"	150#	F751	2"	150#	F201
3/4"	300#	F753	2"	300#	F203
3/4"	600#	F756	2"	600#	F206
1"	150#	F101	3"	150#	F301
1"	300#	F103	3"	300#	F303
1"	600#	F106	3"	600#	F306

FMLG-WP Welded Pad Type Level Indicator

Direct Weld-On Gauge

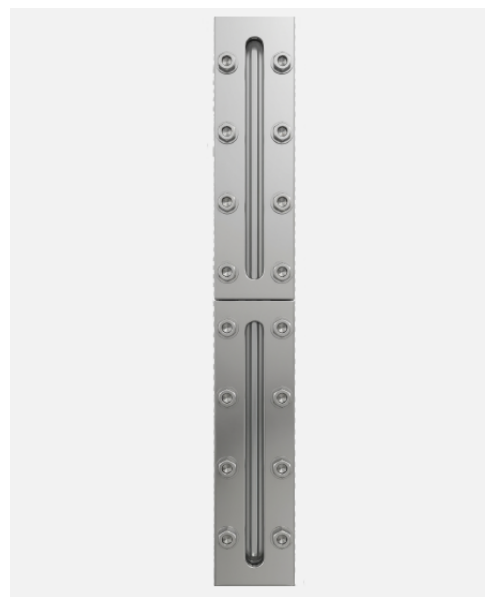
Welded pad type level indicators are used where conventional gauge and valve construction is impractical because of solids in suspension, where thermal error from piping to a gauge glass cannot be tolerated, or where space is limited. The unit is welded directly onto the vessel wall, flat or contoured to the vessel radius.

General Features

- Welds directly onto the tank or vessel; no piping-induced thermal error.
- Standard flat-bottom construction; contoured to vessel radius on request.
- Available in single or multiple (stacked) sections for extended range.

Industries & Applications

- Specialty Chemicals: High-pressure crystallization reactors.
- Chemical Processing: Agitated chemical mixing vessels.
- Chemical / Food Processing: High-viscosity slurry holding tanks.
- Mining / Minerals Processing: Applications with solids in suspension.
- Process Engineering: Vessels where piping-induced thermal error to a remote gauge glass cannot be tolerated.



Key Specifications

Body Material	Carbon Steel or AISI 316 SS
Max. Pressure (SS)	40 kg/cm ²
Max. Temperature (SS, TFE)	-40C to 240C
Mounting	Direct weld-on

General Specifications — FMLG-WP (Direct Weld-On Gauge)

Parameter	Specification
Model	FMLG-WP
Mounting Type	Direct weld-on
Process Connection	Directly welded flat pad construction
Components	Body, retainer caps, lens, lens cushion gaskets, O-rings/packing, retainer cap bolts, set screws

Gauge Classification (Body / Gasket / Pressure / Temperature)

Body Material	Gasket Material	Max. Pressure	Max. Temperature
Carbon Steel	EPDM	20 kg/cm ²	-25C to 150C
Carbon Steel	TFE	20 kg/cm ²	-25C to 240C
Carbon Steel	Viton	20 kg/cm ²	-25C to 205C

Body Material	Gasket Material	Max. Pressure	Max. Temperature
Carbon Steel	Silicone	20 kg/cm ²	-25C to 240C
AISI 316 SS	EPDM	40 kg/cm ²	-50C to 150C
AISI 316 SS	TFE	40 kg/cm ²	-40C to 240C
AISI 316 SS	Viton	40 kg/cm ²	-25C to 205C
AISI 316 SS	Silicone	40 kg/cm ²	-55C to 240C

Ordering Code

Length/section is itemised as a selectable code below so a custom weld-on length can be fully specified from this document alone.

Segment	Parameter	Options	Digit / Code
1	Process Connection	Welded	W
2	Body Material	Carbon Steel	CS
		AISI 316 SS	6T
3	Retainer Cap	Carbon Steel	S1
		AISI 316 SS	6T
4	Sight Glass	Borosilicate (max. 10 kg/cm ² , 120C)	BO
		Ilmadur (max. 20 kg/cm ² , 120C)	IL
		Klinger (max. 40 kg/cm ² , 240C)	KL
5	Gasket	Teflon	TF
		Graphoil	GR
		Gylon	GY
		Blue-Gard	BG
		Non Standard	XX
6	O-Ring	Viton	VT
		Teflon	TF
		Graphite	GR
		Buna N	BN
		Silicone	SI
		EPDM	EP
		Non Standard	XX
7	Retainer Cap Bolts & Set Screws	Carbon Steel	CS
		Stainless Steel	SS
8	Length / Sections	Specify visual length in mm, single section	L####-1
		Specify visual length in mm per section, multi-section stacked	L####-n

Ordering example (illustrative): FMLG-WP-W-6T-6T-KL-TF-VT-SS-L1200-1

Modifications may take place and materials specified may be replaced without prior notice. Confirm all fields before placing an order.